

Shuyue Jia (Bruce)

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RESEARCH STATEMENT

I am an AI researcher with production engineering experience, focusing on both fundamental and applied AI. Please visit my Google Scholar (**with >900 citations, >10 H-index**) to explore my recent publications.

- **Text/Image/Video Processing, Understanding, Generation, and Quality Assessment**
- **AI for Robotics and Large World Model (LWM)**
- **Multimodal Foundation Model (FM) and Large Vision-language Model (LVLM)**
- **Autonomous AI System, AI Agent, and Multi-agent System (MAS)**
- **AI for Human Health**
- **EEG-based Brain-computer Interface (BCI)**

I am dedicated to advancing frontier AI technologies that effectively and efficiently address real-world challenges. My ultimate goal is to **develop safe, reliable, extensible, and cost-efficient Artificial General Intelligence (AGI) systems for science and medicine**. I actively build AI infrastructure and software that enable reproducible research and accelerate progress across the field. All source code is shared on 🐙 GitHub (**>2.1K stars and >300 forks**). Models and datasets are available on 🤗 Hugging Face (**>730K total downloads and >30K downloads per month**) and 🐳 Docker Hub, and applied research will be developed into 🛠️ **products**. Additionally, I have hands-on experience in **distributed LLM/LVLM training, deployment, and million-scale data processing**.

SERVICES

Chair, IEEE Boston Section Circuits and Systems Society, Boston, MA	2026 - Present
Program Committee Member, Association for Computational Linguistics (ACL) Industry Track, San Diego, CA	2026
Technical Committee Member, Embodied AI Systems, IEEE Systems, Man, and Cybernetics Society	2026 - Present
Track Chair, IEEE MIT Undergraduate Research Technology Conference (URTC), Cambridge, MA	2025 - Present
Speaker Program Chair, IEEE International Conference on AI and Data Analytics (ICAD), Boston, MA	2025 - Present
Scientific Program Committee Member, AMIA Clinical Informatics Conference, Anaheim, CA	2025

AWARDS

Arthur Winston Student Award, IEEE Boston Section, USA	2025
Wiley Top Cited Article 2023-2024, Wiley Top Cited Papers, WILEY, USA	2025
Graduate Fellow: Research Fellowship, Boston University, USA	2024 - 2028
Distinguished Computer Engineering Fellowship, Boston University, USA	2023 - 2024
Research Assistantship, City University of Hong Kong, Hong Kong	2021 - 2023
CityU Top 5 Runner, City University of Hong Kong, Hong Kong	2021
Innovation Scholarship, Northeast Electric Power University, China	2017 - 2019
Outstanding Athlete, Northeast Electric Power University, China	2017
Excellent Student Scholarship, Northeast Electric Power University, China	2016 - 2020
Interdisciplinary Contest In Modeling, USA <i>Honorable Mention</i>	2019
Mathematical Contest In Modeling, Jilin, China <i>First Prize</i>	2018
Chinese Physics Olympiad (CPHO), China <i>Third Prize</i>	2015
National High School Math League, China <i>Second Prize</i>	2015

ACADEMIC APPOINTMENTS

Boston University , Boston, MA, USA	Research Assistant	May 2024 - Present
• Supervisor: Prof. Vijaya B. Kolachalama • Projects: Scientific & Medical Foundation Model, AI Grounding, Specialized Intelligence, and Personal Intelligence		
City University of Hong Kong , Hong Kong, China	Research Assistant	Sept 2021 - Apr 2023
• Supervisor: Prof. Shiqi Wang • Projects: Image Quality Assessment (IQA) and Perceptual Optimization		

EDUCATION

Boston University , Boston, MA, USA	Sept 2023 - May 2028 (expected)
• Ph.D. Candidate in Computer Engineering, Department of Electrical and Computer Engineering • Supervisor: Prof. Vijaya B. Kolachalama • Expertise: Scientific & Medical Multimodal Foundation Model and Medical Imaging with AI	
City University of Hong Kong , Hong Kong, China	May 2021 - June 2023
• M.Phil. in Computer Science, Department of Computer Science • Supervisor: Prof. Shiqi Wang • Expertise: Image Quality Assessment, Perceptual Optimization, and Deep Learning • Thesis: No-reference image quality assessment via non-local modeling	
Northeast Electric Power University , Jilin, Jilin, China	Sept 2016 - June 2020
• B.Eng. in Intelligence Science and Technology, School of Automation Engineering • Supervisor: Prof. Yimin Hou and Prof. Jinglei Lv • Thesis: Brain-computer interface signal classification and its applications based on deep learning methods	
University of California, Irvine , Irvine, CA, USA	Jul - Sept 2017
• Visiting Student, Department of Computer Science	

PUBLICATIONS

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- [1] PodGPT: An audio-augmented large language model for research and education [📄 Paper] [🔗 Code] [📄 Slides]
Shuyue Jia, Subhrangshu Bit, Edward Searls, Meagan Lauber, Lindsey Claus, Pengrui Fan, Varuna Jasodanand, Divya Veerapaneni, William Wang, Rhoda Au, Vijaya B. Kolachalama * [😊 Models] [📄 Supplementary Material]
Nature npj Biomedical Innovations
- [2] Agentic memory-augmented retrieval and evidence grounding for medical question-answering tasks [📄 Paper] [🔗 Code]
Shuyue Jia, Subhrangshu Bit, Varuna Jasodanand, Yi Liu, Vijaya B. Kolachalama * [📄 Slides] [😊 Datasets]
International Journal of Medical Informatics (IJMI, IF in 2026: 5.0) [📄 Supplementary Material]
- [3] MedSyn: Text-guided anatomy-aware synthesis of high-fidelity 3D CT images [📄 Paper] [🔗 Code]
Yanwu Xu, Li Sun, Wei Peng, **Shuyue Jia**, Katelyn Morrison, Shyam Visweswaran, Motahhare Eslami, Kayhan Batmanghelich *
IEEE Transactions on Medical Imaging (IEEE T-MI, IF in 2024: 8.9)
- [4] DocAgent: An agentic framework for multi-modal long-context document understanding [📄 Paper] [🔗 Code]
Li Sun, Liu He, **Shuyue Jia**, Yangfan He, Chenyu You *
The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)

- [5] No-reference image quality assessment via non-local dependency modeling  Paper  Code  Slides  Poster
Shuyue Jia, Baoliang Chen, Dingquan Li, Shiqi Wang *
The 24th IEEE International Workshop on Multimedia Signal Processing (IEEE MMSP)
- [6] Learning from mixed datasets: A monotonic image quality assessment model  Paper  Code
 Zhaopeng Feng, Keyang Zhang, **Shuyue Jia**, Baoliang Chen, Shiqi Wang *
Electronics Letters
Awarded Wiley Top Cited Article 2023–2024
- [7] GCNs-Net: A graph convolutional neural network approach for decoding time-resolved EEG motor imagery signals  Paper  Code (**with >1K stars and >200 forks**)  Slides
Shuyue Jia, Yimin Hou, Xiangmin Lun, Ziqian Hao, Yan Shi, Yang Li, Rui Zeng, Jinglei Lv *
IEEE Transactions on Neural Networks and Learning Systems (IEEE T-NNLS)
IF in 2022: 14.255, Top 0.1% most cited articles published in Engineering in 2022
- [8] Deep feature mining via attention-based BiLSTM-GCN for human motor imagery recognition
Shuyue Jia *, Yimin Hou, Xiangmin Lun, Shu Zhang, Tao Chen, Fang Wang, Jinglei Lv  Paper  Code  Slides
Frontiers in Bioengineering and Biotechnology
IF in 2021: 6.064, Top 1% most cited articles published in Engineering in 2022
- [9] A novel approach of decoding EEG four-class motor imagery tasks via scout ESI and CNN
 Yimin Hou, Lu Zhou *, **Shuyue Jia**, Xiangmin Lun  Paper  Code (**with >200 stars and >50 forks**)
Journal of Neural Engineering
IF in 2020: 5.379, Top 1% most cited articles published in Engineering in 2020
- [10] PMU measurements based short-term voltage stability assessment of power systems via deep transfer learning  Paper
 Yang Li *, Shitu Zhang, Yuanzheng Li, Jiting Cao, **Shuyue Jia**
IEEE Transactions on Instrumentation and Measurement (IEEE T-IM)
IF in 2023: 5.6, Top 1% most cited articles published in Engineering in 2023
- [11] Improving Performance: A Collaborative Strategy for the Multi-data Fusion of Electronic Nose and Hyperspectral to Track the Quality Difference of Rice  Paper
 Yan Shi, Hangcheng Yuan, Chenao Xiong, **Shuyue Jia**, Jingjing Liu, Hong Men *
Sensors & Actuators: B. Chemical, IF in 2021: 9.221

Note:

1. The impact factor (IF) corresponds to the journal's impact factor in the year the paper was published.
2. * denotes the Corresponding Author

MANUSCRIPTS UNDER SUBMISSION

- [1] Vision-language framework for multi-sequence brain magnetic resonance imaging  Product  Paper
 Diala Lteif, **Shuyue Jia**, Subhrangshu Bit, Artem Kaliev, Asim Z. Mian, Juan E. Small, Balamurugan Mangaleswaran, Bryan A. Plummer, Sarah A. Bargal, Rhoda Au, Vijaya B. Kolachalama  Supplementary Material
To be submitted
- [2] Gradient-guided adapter merging for neuroimaging vision-language models  Paper

THESIS

- No-reference image quality assessment via non-local modeling [📄 Permanent Link] [📄 Defense Slides]

Shuyue Jia

M.Phil. Thesis, City University of Hong Kong, May 2023.

- Brain-computer interface signal classification and its applications based on deep learning methods

Outstanding Thesis Award [🔗 Related Publication] [📄 Defense Slides]

Shuyue Jia

B.Eng. Thesis, Northeast Electric Power University, May 2020.

SELECTED PROJECTS

Vision-language Framework for Multi-sequence Brain MRIs

2025 - 2026

Mentor: Prof. Vijaya B. Kolachalama, Boston University

- **Dataset scale:** **>850K multi-sequence brain MRIs, >73K patient visits, >1M instruction tuning data**
- **Model scale:** 35B MoE (3B activated), 27B, and 9B
- **Neuroimaging (MRI) data processing & harmonization:** DICOM → NIfTI → 2D/3D/4D volumes → MONAI RAS anatomical orientation → Move through-plane axis (largest voxel spacing) → Intensity normalization → Square padding & resizing → Slices to frames with discarding all-black slices, uint8 scaling, and radiological orientation → Frames to lossless video.
- **Instruction tuning dataset curation:** Introduced 12 question types (as tasks), including 6 classification-based QA types (Choice Verification, Yes/No QA, True/False, Multiple Choice QA, Minimal Pair Discrimination, and Textual Entailment, with answer distribution balanced) and 6 generation-based QA types (Classification, Extraction, Cloze (Masked Word), Factual QA, Fill-in-the-Blank, and Sentence Completion).
- **Model training:** 35B MoE (3B activated): Hugging Face trainer with Flash Attention 2, causal-conv1d kernel, and torch compiled on 4 B200 GPUs (183GB memory, Blackwell, CUDA 13.2), 27B Dense model: Flash Attention 3, causal-conv1d kernel, and torch compiled on 2 H200 NVL GPUs (143GB memory, Hopper, CUDA 12.8), and 9B Dense model: Flash Attention 2, causal-conv1d kernel, and torch compiled on 2 RTX PRO 6000 GPUs (97GB memory, Blackwell, CUDA 13.2).
- **Model evaluation & inference:** vLLM generate engine (0.17.0+cu132) compiled on 2 RTX PRO 6000 GPUs (97GB memory, CUDA 13.2).
- **Model deployment** (served on <https://remind.bu.edu>): 35B MoE (3B activated) with vLLM serve engine (0.24.0) on 4 RTX 3090 GPUs (24GB memory, CUDA 13.2).

Continual Pre-training of LLMs via Podcast Audio, RAG, & Agentic RAG

2024 - 2025

Mentor: Prof. Vijaya B. Kolachalama, Boston University

- **Dataset scale:** **>42M text tokens, >3700 hours of podcasts, >10K top journal articles**
- **Model scale:** 2B, 7B, 11B, 8×7B MoE, 70B
- **Model training & inference:** 12 L40S GPUs (48GB memory, CUDA 12.8) with vLLM generate engine (0.6.2)
- **Computational framework:** Developed a computational framework designed to improve Large Language Models (LLMs) leveraging the up-to-date and informational content of publicly accessible Science, Technology, Engineering, Mathematics, and Medicine (STEMM) podcasts. [📦 Product] [📄 Paper] [🔗 Code]

- **Dataset curation:** Collected and processed Creative Commons Attribution (CC BY) licensed and *The New England Journal of Medicine* (NEJM) podcasts from scratch.
- **Model training & evaluation & inference:** Created and maintained  **GitHub Library PodGPT**, a whole pipeline that consists of Continual Pre-training, Low-Rank Adaptation (LoRA) for Efficient Fine-Tuning, GPTQ Model Quantization (**with >30K downloads per month on  Hugging Face**), STEMM Benchmarking, and real-world deployment by vLLM serve engine.
- **RAG pipeline for AI grounding:** Introduced and maintained  **PodRAG**, a **Retrieval-Augmented Generation (RAG)** pipeline featuring retrieval and reranking using PubMed Central and NEJM articles, designed to reduce hallucinations and provide grounding through fact-checking.
- **April 2025 update:** We extended this work to a unified, open-source 72B LLM-based agentic system (served on 4 L40S GPUs, vLLM 0.6.3 serve, openai[aihttp]-compatible server), integrating document retrieval, re-ranking, evidence grounding, and diagnosis generation to support dynamic, multi-step medical reasoning.
Updated dataset scale: **>30M documents, >68M snippets**  **Paper**  **Code**
- **April 2026 update:** We extended PodGPT's evaluation on Nephrology Self-Assessment Program (NephSAP) multiple-choice QA dataset with clinical validations by 3 board-certified nephrologists (5 questions $\times$ 8 nephrology subject areas) to better understand its capabilities and limitations within this specialized clinical domain.  **Paper**  **Inference Code**  **Figures Code**

Non-local Modeling for Image Quality Assessment

2021 - 2023

Mentor: Prof. Shiqi Wang, City University of Hong Kong

- Built a **Superpixel-based Graph Neural Network (GNN)** model to explore the non-local interactions in quality prediction.  **Paper**  **Code**
- Presented the definitions of **Global Distortions** and **Local Distortions** for IQA.  **Thesis**  **Slides**

Graph Representation Learning for EEG-based BCI Classification

2019 - 2021

Mentor: Prof. Yimin Hou, Northeast Electric Power University and Prof. Jinglei Lv, The University of Sydney

- Implemented a **Convolutional Neural Network (CNN)** architecture with Morlet wavelet joint time-frequency analysis for EEG time series.  **Paper**  **Code** (**with >200 stars and >50 forks**)
- Introduced **Graph Modeling** and **Graph Representation Learning** to decode raw EEG signals by cooperating with the functional topological relationship of electrodes.  **Paper**  **Code**
- Proposed an accurate and responsive EEG signal recognition system via **Bidirectional Long Short-term Memory (BiLSTM)** with the **Attention Mechanism** and **Graph Neural Networks**.  **Paper**  **Code**
- Created and maintained  **GitHub Library EEG-DL** (**with >1K stars and >200 forks**), a Deep Learning library written by TensorFlow for EEG signal classification.

Statistical Measures for Sonar Image Segmentation

2018 - 2019

Mentor: Prof. Shengxi Jiao, Northeast Electric Power University

- Engineered an image-processing pipeline to enhance ship-to-background contrast, comprising **Image Filtering** using discrete cosine transform (DCT), **Edge Detection** using Roberts cross operator, **Object Localization** through thresholding of a one-dimensional histogram, and **Morphological Local Enhancement** using a dilation operator.
- Improved segmentation accuracy from 80% to 86% by incorporating the preprocessing stage.
- A patent was granted based on this project, and the  source code is publicly available.

INDUSTRY EXPERIENCE

Philips Research, Shanghai, China

Natural Language Processing Intern

Jul - Oct 2020

Mentor: Dr. Shuang Zhou, Precision Diagnosis & Image Guided Therapy (PD&IGT) Department.

- **Medical Concept Mapping:** Developed a three-level framework to map user queries to standardized medical terms: **Byte-Pair Encoding** and **FMM & BMM Algorithms** for subword generation and matching at the syntactic level, **Word2Vec-Based Cosine Similarity** for semantic matching, and a **Knowledge Graph** for pragmatic matching.
- **Medical Named Entity Recognition (NER):** Evaluated and compared multiple methods, including CRF++, a character-level BiLSTM with CRF, and hierarchical character-level and word-level BiLSTM/CNN models with CRF, and deployed the trained model with a web app using Python Flask and Docker (**with 400+ downloads on  Docker Hub**).
- **Dynamic Webs Crawling:** **crawled >620K pages** from the China National Science and Technology Library (NSTL) via Python threading.

TEACHING APPOINTMENTS

EC327: Introduction to Software Engineering, Boston University, MA, USA

Spring 2025

- Introduction: This course aims to introduce students to software design, object-oriented programming techniques, elementary data structures and algorithmic analysis, and software engineering principles.
- Responsibility: Prepared lab session materials, led lab sessions for students, and designed homework assignments with automated grading.

SOFTWARE, LIBRARY, AND TOOLKIT

Software - **ReMIND** [ Product] [ Paper]

- An online platform for deploying our latest video foundation model for multi-sequence brain MRIs.

Software - **PodGPT** [ Product] [ Paper] [ Code]

- An online platform for deploying our latest multimodal foundation models for medical and clinical applications.
- The success of the PodGPT project is a result of **collaborative teamwork** by a group of members of the Kolachalama Laboratory. The PodGPT Library is maintained by the Kolachalama Laboratory at Boston University.

Library - **PodGPT Library** [ Paper] [ Code]

- A library for benchmarking multilingual medical Large Language Models (LLMs).

Library - **EEG Deep Learning Library** (**with >1K stars and >200 forks**) [ Code]

- A Deep Learning (DL) library written by TensorFlow for EEG tasks (signals) classification.

Toolkit - **PromptCraft** [ Code] [PyPI Package]

- A prompt perturbation toolkit from the character, word, and sentence levels for prompt robustness analysis.
- Install: `pip install promptcraft`

INVITED REVIEWER

Conducted over 200 technical and multidisciplinary paper reviews for top-tier journals and computer science conferences, contributing to the advancement of research and maintaining high publication standards.

IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)

IEEE Transactions on Multimedia (T-MM)

IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT)

IEEE Transactions on Neural Networks and Learning Systems (T-NNLS)

IEEE Transactions on Medical Imaging (T-MI)

IEEE Transactions on Industrial Informatics (T-II)
 IEEE Transactions on Biomedical Engineering (T-BME)
 IEEE Transactions on Human-Machine Systems (T-HMS)
 IEEE Transactions on Affective Computing (T-AFFC)
 IEEE Transactions on Cognitive and Developmental Systems (T-CDS)
 IEEE Transactions on Emerging Topics in Computational Intelligence (T-ETCI)
 IEEE Journal of Biomedical and Health Informatics (J-BHI)
 IEEE Journal of Selected Topics in Signal Processing (J-STSP)
 IEEE Open Journal of the Industrial Electronics Society (OJ-IES)
 IEEE Open Journal of the Computer Society (OJ-CS)
 IEEE MultiMedia (MM)
 IEEE Sensors Journal
 Computer Vision and Image Understanding (CVIU)
 Journal of Medical Internet Research (JMIR)
 IET Image Processing
 IET Science, Measurement & Technology
 Neural Information Processing Systems (NeurIPS) 2026
 Association for Computational Linguistics (ACL) Industry Track 2026
 International Conference on Learning Representations (ICLR) 2025, 2026

STEERING COMMITTEE MEMBER

2027 IEEE International Conference on AI and Data Analytics (IEEE ICAD)	June 2027
Venue: TBD	
2026 IEEE International Conference on AI and Data Analytics (IEEE ICAD)	June 2026
Venue: Fulton Hall & Stokes Hall, Boston College, 140 Commonwealth Ave., Chestnut Hill, MA, USA	
2025 IEEE International Conference on AI and Data Analytics (IEEE ICAD)	June 2025
Venue: Joyce Cummings Center, Tufts University, 177 College Ave., Medford, MA, USA	

SKILLS

Coding: Python, C/C++, MATLAB, R, HTML, CSS
Library: TensorFlow, PyTorch, Scikit-learn, Hugging Face (Transformers, Accelerate, PEFT, and Datasets), vLLM, DeepSpeed ZeRO, FlashAttention, Causal Convld, and GPTQModel
Tools: Git, SVN, Unix Shell, Vim, Markdown, L^AT_EX, Docker, K8s
Hardware: NVIDIA Jetson Nano (Ubuntu), Raspberry Pi (Raspbian OS), 80C52 Microcontroller (Embedded C language)
Photography: Bruce S. JIA Photography